

RELATED WORK · FAIR-TRACE / SCAFF

How five papers evaluate fairness in LLM recommenders and agents

FaiRLLM · CFaiRLLM · SCOPED-Hiring · AgentFairBench · Clinical instability floor

Five papers at a glance

Paper	Setting	What is swapped	Unit measured	Headline
FaiRLLM Zhang et al., RecSys 2023	ChatGPT recommends music and movies	Sensitive attribute stated in the prompt vs. a neutral prompt	Top-20 list	Lists shift by attribute; race (movies) and religion (music) worst
CFaiRLLM Deldjoo & Di Noia, TIST 2025	LLM recommends from real user histories	Single and intersectional attributes added to a user profile	Top-K list vs. true preferences	Measured unfairness depends on the reference and profile sampling
SCOPED-Hiring Zhao et al., arXiv 2026	Two-stage multi-agent hiring committee	One resume cue: name, age, country, school tier, career gap	Full decision trajectory	Balanced hire rates hide unequal process; repair cuts burden 72%
AgentFairBench Morla et al., arXiv 2026	Agents in hiring, lending, triage	Name-coded race × gender in matched profile sets	Action, score, info request	No effect above an arity-matched noise floor (Claude Haiku 4.5)
Instability floor Bellibatlu et al., arXiv 2026	Six-stage clinical agent trajectory	Race, sex, age, insurance, English proficiency	Each clinical action	Identical reruns flip 8.7% of actions; no gap beats that floor

FaiRLLM: Is ChatGPT fair for recommendation?

Question Does ChatGPT change its recommendations when the user states a sensitive attribute?

SETUP

- Music (500 popular singers) and movies (500 top directors); request $K = 20$ titles
- 8 attributes: age, country, gender, continent, occupation, race, religion, physical traits
- Neutral prompt “I am a fan of X” vs. “I am a/an [attribute] fan of X”; greedy decoding

METRICS

- Similarity of each attribute's list to the neutral list: Jaccard, SERP*, PRAG*
- Unfairness = spread across values: SNSR (max – min) and SNSV (std)

FINDINGS

- Most attributes shift the lists; race is worst in movies (PRAG* SNSR 0.22)
- “African” and “Asian” users drift furthest from the neutral list
- Holds with typos near the attribute word and with Chinese-language prompts

HOW THEY EVALUATE

- Reference = the neutral prompt's output
- Any difference counts as unfairness, even if harmless
- Output-level only; one model; no noise floor (greedy decoding)
- For SCAFF: this is D^{nat} on a single-stage system

CFaiRLLM: consumer fairness against true preferences

Question Is a change in recommendations unfair, or just personalization?

SETUP

- Real histories: MovieLens-1M and LastFM-1K, about 150 users each
- Profile = 10 items sampled three ways: random, top-rated, most recent
- Attribute added to the prompt, alone or intersectional (e.g., age × gender)

METRICS

- Item similarity to the neutral list (Jaccard, PRAG), as in FaiRLLM
- True-preference alignment: recommendations scored against held-out user items
- Same summaries across groups: SNSR and SNSV

FINDINGS

- Similarity-based measures report more unfairness than true-preference alignment
- Intersectional groups show larger gaps; profile sampling changes the conclusion

HOW THEY EVALUATE

- Fixes FaiRLLM's flaw: judge differences against the user's real preferences
- The unfairness number depends on choices the evaluator makes
- Still output-level; no stage attribution
- For SCAFF: mirrors our split of sensitivity (D^A) from quality

SCOPED-Hiring: process-aware fairness diagnosis

Question Can balanced final outcomes hide unfair treatment inside a multi-agent process?

SETUP

- Committee: 3 screeners → 3 executives; debate, votes, private assessments logged
- Resume variants from real CVs change one cue, merit fixed; 311K trajectories, 3 LLMs

METRICS · SIX LENSES

- Outcome · counterfactual (flip rate, score ATE) · process (investigation requests, bias language)
- Pathway (Stage-1 pass gap vs. Stage-2 conversion gap) · debate dynamics · design effects

FINDINGS

- Process-level signals 2.7–4.7× larger than outcome-level signals
- Career gaps trigger suspicion; identity cues trigger more investigation
- Targeted prompt repair: −72.3% burden, hire rate moves 1.86 pp; generic reminder fails

HOW THEY EVALUATE

- Shows where a gap appears along the pipeline
- Never swaps upstream state, so it can't tell caused from inherited
- No planted ground truth; validated by construct checks and repair
- For SCAFF: closest neighbor; our crossover is the difference

AgentFairBench: do LLM agents discriminate when they act?

Question Do agents treat matched profiles differently in actions, not just answers?

SETUP

- Hiring, lending, medical triage; synthetic profiles in matched sets
- Only a name-coded race × gender signal varies; 4 scaffolds (direct, CoT, panel, info request)

METRICS

- Flip rate, score spread, action-rate gap, info-request gap
- Bootstrap CIs, permutation tests, false-discovery-rate control; test-retest reruns

FINDINGS

- A six-group range vs. a two-run noise gap overstates disparity $\approx 2.25\times$ by arity alone
- Against an arity-matched floor: 0/120 pairwise, 0/9 omnibus contrasts (Claude Haiku 4.5)
- A planted-bias test confirms detection (construction barely described)

HOW THEY EVALUATE

- Noise floor built from reruns of identical inputs
- Compare like with like: same number of groups in signal and null
- Planted bias checks power
- For SCAFF: already cited by Yashar; borrow the arity-matched null

Counterfactual audits need a per-action instability floor

Question What does a counterfactual flip rate mean if the agent flips on identical input?

SETUP

- FairMedAgent: synthetic clinical vignettes, six-stage trajectory (five model decisions)
- Descriptors: race, sex, age, insurance, English proficiency, and intersections

FLOOR PROTOCOL

- Rerun an identical condition 10× over 16 vignettes; count action changes per cell
- Repeat on a second model; test majority vote over 5 draws

FINDINGS

- 8.7% of actions change with nothing varied; 0.022 (ICU escalation) to 0.179 (substance caution)
- Second model: 6.7% floor, same ranking of actions (Spearman 0.94)
- Majority vote removes 39% of the floor, then flattens; no disparity beats the floor

HOW THEY EVALUATE

- Noise is per action, not one global number
- Report the floor next to every flip rate
- Replication helps but doesn't remove the floor
- For SCAFF: our qwen reruns disagree on 6.4% of stage decisions

What each evaluation checks

	Significance	Check prefs.	bycat ages	Diherf test	granted with	Noise	Test plan
FaiRLLM	✓	✓	—	—	—	—	—
CFaiRLLM	✓	—	✓	—	—	partial	✓
SCOPED-Hiring	✓	—	—	—	✓	✓	—
AgentFairBench	✓	partial	partial	—	—	✓	—
Instability floor	✓	✓	✓	✓	✓	partial	✓
SCAFF (ours)							

Partial: SCOPED-Hiring runs significance tests but no rerun floor. The instability-floor paper's clinician-adjudicated reference is still in progress. SCAFF computes within-condition noise but does not yet use it as the D^A threshold.

What to borrow for SCAFF's evaluation

Threshold D^A against each stage's own noise

Replace the fixed 0.10 with a per-stage rerun floor, arity-matched. Our qwen noise (6.4%) sits right beside their 6.7–8.7%.

Keep sensitivity and consequence separate

CFaiRLLM shows a raw difference overstates unfairness. Report quality against task truth next to D^A .

Make planted ground truth a headline

Only AgentFairBench plants bias, and barely describes it. Report localization vs. chance (1/6), per-stage recall, and the confusion matrix.

Position against SCOPED-Hiring

They show where a gap appears. SCAFF's crossover shows whether a stage causes it (D^A) or inherits it (D^Z).

Use repair as validation

Their targeted vs. generic repair matches our true-stage vs. other-stage removability (toy: 0.58 vs. 0.18).

SOURCES

arXiv 2305.07609 · 2403.05668
arXiv 2609.02092 · 2606.16723
arXiv 2609.03221